Parul Singh

AI/ML | NLP | LLMs | RAG | DATA ENGINEERING | DATA SCIENCE

+91-7007801182 | parulsingh1074@gmail.com | GitHub | Portfolio | LinkedIn

PROFILE SUMMARY

Computer Science graduate and AI/ML Intern with hands-on experience building Python applications, data-processing and synthetic-data pipelines, Django automation systems, and LLM/RAG solutions. Skilled in data validation, statistical checks, semantic retrieval, model workflows, and debugging multi-step AI-agent trajectories. Experienced in translating specifications into working applications and automated workflows. Seeking Python Developer, AI/ML Engineer, Data Engineer, or Data Science opportunities that value analytical problem-solving and reliable implementation.

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript, Object-Oriented Programming
- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, Generative AI, RAG
- **Frameworks & Libraries:** TensorFlow, Keras, Scikit-learn, LangChain, FAISS, Django, Flask, Streamlit
- **Data & Analytics:** Pandas, NumPy, SciPy, Matplotlib, NLTK, Feature Engineering, Model Evaluation, Statistical Validation, Synthetic Data Generation
- **Testing & Data Quality:** Pytest, automated validation, data integrity checks, range/distribution/correlation checks, bounded repair
- **Web, Tools & Databases:** React, Tailwind CSS, Docker, Git, GitHub, Vercel, Netlify, Render, MySQL, Supabase/PostgreSQL, Claude API

EXPERIENCE

FAC Infosec Private Limited — Remote

Artificial Intelligence & Machine Learning Intern — Aug 2026 – Present

- Collaborating in a team of three to develop a LangGraph-based runtime for autonomous software-engineering tasks, with structured investigation, planning, implementation, and verification.
- Building two-tier validation using AST/Python syntax checks, file-scope enforcement, repository tests, and an LLM-based semantic gate before accepting code changes.
- Implementing isolated Git worktrees, bounded retry/strike limits, SQLite checkpointing, and failure analysis to improve recovery and execution traceability.

Indian Institute of Technology, Kanpur — IITK Campus, Kanpur

Research Intern · SURGE Program — May 2024 – Jul 2024

- Developed a Django-based automation system integrating regulatory sources, eliminating manual document search and saving approximately 20 hours per week.
- Built web-scraping pipelines to fetch, process, and store regulatory documents from 6+ Electricity Regulatory Commission websites.
- Designed automated update/comparison workflows and collaborated with a team to deliver a practical regulatory tracking solution.

PROJECTS

Jev State Engine | Python, LangGraph, Git Worktrees, AST, SQLite, Ollama (Qwen 2.5 Coder), Gemini API, Pytest | GitHub

• Built a deterministic state-machine runtime for autonomous coding tasks, using mechanical AST/pytest checks and an LLM semantic gate to control untrusted code-generation agents.
• Isolated code changes in Git worktrees and added a 3-strike circuit breaker with forensic post-mortems using a local Qwen 1.5B advisor.
• Benchmarked 17 runs on sqlparse and humanize; tasks completed in one subgoal at ~9.5k tokens, with 100% test integrity and ~75% lower token overhead versus standard ReAct agents.

Synthetic Data Generator | Python, Pandas, NumPy, SciPy, Streamlit, Claude API, Pytest | GitHub

- Built a workflow that converts Excel specifications into synthetic datasets and exports CSV, XLSX, JSON, DOCX, and PDF deliverables.
- Added checks for row counts, unique IDs, ranges, distributions, and correlations, with bounded repair and a 60+ test suite.
- Included offline mock mode and safe formula evaluation to support repeatable generation and testing without a live API dependency.

QueryPDF – AI-Powered Document Assistant | LLMs, Embeddings, Vector Search, LangChain | GitHub

- Built a document assistant supporting natural-language questions about PDF content.
- Implemented semantic retrieval using embeddings and vector search to find relevant document passages.
- Integrated an LLM with retrieved context to generate context-aware answers.

Solar Panel Defect Classification | Python, TensorFlow, EfficientNet, Streamlit | GitHub

- Developed an EfficientNet-based deep-learning model to classify solar-panel images across six defect categories.
- Applied image preprocessing and augmentation to prepare image inputs and support model training.
- Built a Streamlit interface for image upload and automated defect prediction.

The Glitch Room | React, Tailwind CSS, Framer Motion, Supabase, Generative AI | Live

- Developed a gamified learning platform with AI-generated challenges and automated feedback.
- Built interactive dashboards, leaderboards, and collaborative Creator Rooms using React and Supabase.
- Added AI-assisted evaluation and progress workflows to support hands-on practice and learning consistency.

EDUCATION

B.Tech, Computer Science & Engineering — Chhatrapati Shahu Ji Maharaj University, Kanpur (CPI: 8.62/10) | 2021–2025

ACHIEVEMENTS

- Selected for IIT Kanpur's SURGE Program 2024; awarded the Institute-Funded Category (stipend recipient).
- Presented The Glitch Room at Unstop Playstorm and PitchFest; selected for Round 1 of Unstop Playstorm.